MIWB BRAZIL 2027 · TAILINGS & MINE CLOSURE

Find it. Recover it. Clean it up.

Satellite detection and mercury-free reprocessing of legacy artisanal gold mining tailings

Caelo Technologies Inc.
Detect and rank legacy sites

AuroClear Technologies Inc.
Reprocess, remove mercury, reclaim

George McDonald, Founder · Canada

THE PROBLEM

Decades of mercury, left in the tailings

2,000+ t

mercury released every year by artisanal and small-scale gold mining

20M

miners in 80+ countries, producing about 20% of the world’s gold

100M

people directly or indirectly exposed to mercury from the sector

Much of it sits in abandoned tailings ponds, where it can methylate and move into fish and the communities that eat them. Nobody has an inventory of where those ponds are.

Source: UNEP planetGOLD; UNEP / GEF GOLD · 2

BUSINESS MODEL

A cleanup that pays for itself

1 · Caelo

Find

Satellites and machine learning rank legacy ponds

2 · Field

Confirm

Sampling checks the feedstock before anyone mobilizes

3 · AuroClear

Recover

Mercury-free reprocessing recovers the gold left behind

4 · AuroClear

Remediate

That gold pays to capture mercury and reclaim the ground

5 · Caelo

Verify

Will confirm from orbit that the cleanup held (2027)

The gold left behind by mercury mining pays to remove the mercury. Remediation is funded by what it recovers, not by grants alone, so it can scale site by site.

CAELO TECHNOLOGIES

Finding legacy tailings from space

1

Detect

Sentinel-2 imagery and machine learning flag tailings by their mineral signature

2

Filter

Sentinel-1 radar, elevation and slope separate quiet legacy ponds from rivers and active ground

3

Rank

Sites are prioritized for field characterization and remediation

Built for remediation planning, not enforcement. Outputs indicate likely tailings; they never claim contaminant concentrations.

PROOF OF CONCEPT

Validated on Canadian ground truth

0.87

F1 score, leave-one-site-out spatial cross-validation

Beats published benchmarks of about 0.7 for tailings classification

Tested on unseen sites, so the score reflects new ground, not memorized locations

Physically interpretable: the strongest features are mineral and iron-oxide signatures

Nova Scotia labelled historic tailings dataset · F1 range 0.869–0.887 · 5

TRANSFER TO THE TROPICS

Applied in Guyana, the harder case

Dense canopy, constant cloud

Small workings under rainforest. A model that holds here should transfer almost anywhere.

Two mining districts scanned

1,400+ candidate water bodies screened, with radar confirmation to remove rivers and wet ground.

Predictions on record

Expected results were committed before any held-out data or field work, so nothing is fitted after the fact.

FIELD VALIDATION · NOVEMBER 2026

Satellite predictions, checked on the ground

17

GPS-logged, lab-assayed samples

Pond sediment, apron and background reference points, plus a blind duplicate for lab QC

Collected with the University of Guyana; full chain of custody to an accredited commercial lab

Positional accuracy recorded at every site

Results due January 2027, before the summit. Field-validated detection in tropical canopy is rare in the literature.

> **Next: a learning loop.** Each campaign will target the sites where the model is least certain, so accuracy improves fastest per sample and new regions can be entered at low cost.

AUROCLEAR TECHNOLOGIES

The gold pays for the cleanup

Reprocess

Legacy pond sediments run through proven gravity equipment, configured and conditioned for this feedstock. No mercury.

Remove mercury

Legacy mercury reporting to concentrate is captured by retort. The full hotspot design is set from field data.

Reclaim

Progressive reclamation from day one, to a background-referenced endpoint, not left to closure.

Pilot plant in Guyana · commissioning target August 2027 · all equipment commercially proven

NEW TECHNOLOGY

Pre-optimized feedstock

CAELO

Selected from orbit

Ponds are ranked before anyone mobilizes, so crews and equipment go only where field checks confirm legacy sediment worth treating.

AUROCLEAR

Conditioned before recovery

A proprietary pre-conditioning step is designed to break up the clay that traps fine gold in pond sediment, before it reaches gravity concentrators.

The equipment is proven. What’s new is preparing the feedstock, choosing it and conditioning it, so proven machines work on material they were never designed for.

PHASE 2 · STATED INTENT

Stopping new mercury at the sluice

Process miners’ sluice discharge and concentrate without mercury.

AuroClear finances the upgrades at no cost to the miner. Better recovery is shared, so the miner comes out ahead and has no reason to use mercury.

Mined ground is reclaimed afterward, with miners taking part.

Why it works

Behaviour changes through economics, not persuasion. It works with existing gold buyers rather than against them.

A two-year development program, contingent on the pilot proving the economics.

COMMUNITY COMMITMENT

5%

Written into the company, not just promised

We are committed to amending AuroClear’s corporate articles to bind 5% of net profits to the communities where we operate, directed by them through existing local structures, with the amount published every year.

INDUSTRY CHALLENGES

How we answer the sector’s pressures

PRESSURE	OUR ANSWER
Water and tailings	Legacy ponds found, reprocessed and reclaimed; process water reused on site
Safety	No open burning of amalgam; legacy mercury captured in a closed retort; fewer field trips into remote terrain
Digitalization and AI	Satellites and machine learning replace district-wide field reconnaissance
Sustainability	Gold-funded cleanup, progressive reclamation, 5% community commitment
Critical minerals	Roadmap: detection extended to artisanal cobalt and lithium workings

MARKET

Legacy tailings are a global problem

80+ countries

South America, West and East Africa, Southeast Asia: wherever mercury amalgamation ran

Policy demand

Minamata Convention parties with significant artisanal mining need national plans to reduce and eliminate mercury

Developed-world sites

Thousands of orphaned and abandoned mines in Canada, Australia and the US: the same detect-and-prioritize problem

Beachhead: Guyana

Prove it here, then expand across the Guiana Shield and Latin America

Source: UNEP planetGOLD; Minamata Convention, Article 7 · 13

ROADMAP

One platform, more of the tailings problem

2027 ROADMAP

Remediation verification

Will track revegetation, water and ground stability to show a cleanup held

2027 ROADMAP

Downstream water

Will map sediment plumes below legacy ponds, flagging where tailings may reach rivers

PLATFORM POTENTIAL

Tailings stability

InSAR ground-movement watch on small ponds that nobody monitors today

PLATFORM POTENTIAL

Critical minerals

Extending detection to artisanal cobalt and lithium workings

Same satellites, same pipeline, same rule: every layer serves remediation, never enforcement.

TEAM & PARTNERS

Built with people on the ground

George McDonald

Founder of Caelo and AuroClear. Former Canadian Armed Forces marine engineer and ship’s rescue diver; biochemistry studies at the University of Guelph; construction contractor.

Field & research

University of Guyana field collaboration; accredited commercial assay laboratory

Network

Member, Canada–Guyana Chamber of Commerce; partner company AuroClear Technologies Inc.

THE ASK

Who we want to meet in Belo Horizonte

Pilot partners

Operators, governments and NGOs with legacy tailings to inventory or remediate

Equipment & engineering

Gravity equipment makers and integrators open to vendor financing

Capital

$75K CAD to build Caelo into a deployable product; impact investors for the AuroClear pilot

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